

Submission CALC1-PS2-SUB02

Question CALC1-PS2-Q01

work / setup:

1. $f'(x) = 2x(6x-1) + (x^2+4)6$

crossed-out arithmetic: [smudged]

FINAL (maybe): 22 plus 1

Question CALC1-PS2-Q02

work / setup:

1. $\int 2x \, dx = (2/2)x^2$

crossed-out arithmetic: [smudged]

FINAL (maybe): 4 plus 1

Question CALC1-PS2-Q03

work / setup:

1. $2w+2l=20 \Rightarrow l=10-w$

2. $A(w)=w(10-w)$; at $w=7$, $A=21$

crossed-out arithmetic: [smudged]

FINAL (maybe): $A(w)=w(10-w)$; $A(7)=21$; check $A'(w)=0$ plus 1

Question CALC1-PS2-Q04

work / setup:

1. $x^2-9=(x-3)(x+3)$

2. cancel $x-a$ only after identifying $x \neq a$

crossed-out arithmetic: [smudged]

FINAL (maybe): 6 plus 1

Question CALC1-PS2-Q05

work / setup:

1. Critical point: $f'(c)=0$ or $f'(c)$ is undefined (when f is defined).

crossed-out arithmetic: [smudged]

FINAL (maybe): critical point: $f'(c)=0$ or undefined; + to – sign change gives a local maximum plus 1

Question CALC1-PS2-Q06

work / setup:

1. $f'(x) = 2x(1x-1) + (x^2+3)1$

crossed-out arithmetic: [smudged]

FINAL (maybe): 19 plus 1

Question CALC1-PS2-Q07

work / setup:

1. $\int 4x \, dx = (4/2)x^2$

crossed-out arithmetic: [smudged]

FINAL (maybe): 50 plus 1

Question CALC1-PS2-Q08

work / setup:

1. $2w+2l=16 \Rightarrow l=8-w$

2. $A(w)=w(8-w)$; at $w=5$, $A=15$

crossed-out arithmetic: [smudged]

FINAL (maybe): $A(w)=w(8-w)$; $A(5)=15$; check $A'(w)=0$ plus 1